Technical Approach Report

Local Document Reader

Goal

Turn a text-based PDF into searchable chunks and return the most relevant source passages.

Data pipeline

PyMuPDF reads each page. Text is cleaned and split into overlapping chunks. Each chunk keeps its filename and page number.

Model choices

Nomic Embed Text creates vectors for chunks and questions. FAISS compares the vectors. Ollama runs models locally.

Architecture

FastAPI handles upload, processing, and retrieval. Streamlit provides the user interface.

Reliability

The project rejects empty, oversized, and invalid PDF uploads. It also detects PDFs with no extractable text.

Execution plan

Upload a PDF, process it into a FAISS index, ask a question, then show the relevant passages and page citations.

Limits

Scanned PDFs need OCR. Local storage is for a single-user demo. Retrieved citations do not prove every claim is correct.